

Ram Ravi

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SUMMARY

Machine Learning Engineer with an M.S. in Data Science and 8+ years building high-throughput, scalable data systems across Morgan Stanley, Athena Capital Research, and Exelon. Applied work spans NLP/RAG, workflow automation, computer vision, and reinforcement learning, with a track record of collaborating across research, engineering and business teams to take ideas from concept through development and deployment.

EDUCATION

Drexel University

Philadelphia, PA

Master of Science in Data Science, GPA: 4.0/4.0

June 2025

- Post-Baccalaureate Certificate in Computational Artificial Intelligence & Machine Learning
- Relevant Coursework: Machine Learning, Deep Learning, Reinforcement Learning, Artificial Intelligence, Cloud Computing, Data Structures and Algorithms

The Hong Kong University of Science and Technology

Hong Kong

Bachelor of Engineering in Electronic Engineering

June 2014

- Dean's List (Spring 2013, 2014)

TECHNICAL SKILLS

Languages: Python, C#, Java, C, SQL, R

Libraries: Pandas, NumPy, Scikit-Learn, PyTorch, PySpark, Matplotlib, LangChain, YOLO, OpenCV, PIL, Optuna

Analytics & ML: Machine Learning, Deep Learning, Time Series, LLM, Data Cleaning and Visualization, Statistics

Cloud and MLOps: MLFlow

Developer Tools: Git, Mercurial, Unix, Roboflow

EXPERIENCE

Machine Learning Researcher

Sep 2025 – Present

Drexel University (Advisor: Prof. Shahin Jabbari)

Philadelphia, PA

- Collaborating with faculty advisor and research team to investigate data-efficient robustness and reliability of ML models using counterfactual-based approaches, applying **ROAR**, **Lp-bounded robust recourse**, and **Dual CF** methods under limited query budgets
- Designed and ran reproducible ML experiments across SBA and German Credit datasets, achieving up to **0.98+ extracted AUC** on tabular models; analyzed sensitivity to query size, regularization (λ), and perturbation bounds
- Benchmarked **ROAR vs optimal L_∞** approaches, identifying extraction fidelity trade-offs between shallow and deep models - finding strong performance on linear models while deep models showed reduced fidelity under constrained query budgets

Exelon (Baltimore Gas & Electric)

Baltimore, MD

Advanced Analysis & Modeling (Co-op)

Jul 2024 – Jan 2025

- Built scalable, high-throughput Python data pipelines to automate forecasting and capacity planning workflows for Distributed Energy Resources (DER) across **1,600+ feeders**, reducing analysis time from **~1 week to under 10 minutes**
- Built distributed grid topology and connectivity models from LoadSEER hierarchy - reconstructing circuit component ordering (reclosers, transformers, fuses) to enable accurate load tracking, prevent overloading, and support outage reconfiguration across BGE's distribution network
- Developed geospatial data pipelines to resolve incomplete DER installation addresses and infer GPS coordinates, improving data quality for downstream systems (IntellioConnect, LoadSEER)

Athena Capital Research

Hong Kong

Senior Trade-Operations Engineer

Sep 2018 – Sep 2023

- Collaborated cross-functionally with quantitative researchers, developers and engineers to support model development and evaluation pipelines, conducting post-trade analytics in Python to identify execution inefficiencies
- Analysed market data across multiple exchanges to surface price return correlations and trading inefficiencies - delivering findings to quantitative researchers as inputs to model refinement

- Developed real-time monitoring tools in Python to detect and flag trading-system errors - reducing investigation and response time from up to **2 hours to under 10 minutes**
- Managed trading risk limits and capital allocations across strategies - ensuring compliance with exchange requirements

Morgan Stanley

Technology Associate / Analyst

New York & Hong Kong

Aug 2014 – Apr 2018

- Built scalable data and automation pipelines supporting risk and trading systems, improving system reliability and reducing manual effort by **10+ hours/week**
- Led migration of Hong Kong's risk account database to Sybase ASE 16.0 - ensuring long-term support and system stability
- Designed and presented a continuous system monitoring framework to senior management, improving production anomaly detection coverage across Institutional Equity systems

PROJECTS

Bookshelf Scanner | YOLOv8, Google Cloud Vision, LangChain, Python 2025

- Built an end-to-end ML pipeline for book discovery integrating object detection, OCR, LLM-based metadata extraction, and embedding-based retrieval
- Trained and evaluated multiple **YOLOv8** variants on a ~20,000-image custom dataset; best model (**YOLOv8x-ORB**) achieved **mAP50 of 93.7%** (precision 86.4%, recall 90.9%)
- Benchmarked OCR engines (**Google Cloud Vision**, Tesseract, Tesseract) using Jaccard and edit-distance metrics; selected Google Cloud Vision (**Jaccard 0.79, ~2× improvement**)
- Designed semantic retrieval system using **Sentence-BERT** embeddings and LLM-generated metadata to enable similarity-based book recommendations

Bipedal Locomotion via Deep RL | PyTorch, Optuna, Gymnasium 2025

- Implemented and tuned **A2C with GAE** from scratch in PyTorch for continuous control, and benchmarked against **PPO and SAC** in Gymnasium's BipedalWalker-v3 environment
- Developed a modular **hyperparameter tuning framework using Optuna** (Bayesian optimization + pruning), running 30+ trials over 13+ hours to optimize key parameters
- Achieved **135.20 mean reward (A2C)** and **321.58 (SAC)** across 50 evaluation trials under a unified training setup
- Analyzed policy generalization by transferring a hardcore-trained SAC agent to the standard environment, achieving a mean reward of **248.02**

Resume Scanner | LangChain, OpenAI 2024

- Built an end-to-end **RAG** pipeline using **LangChain** and **OpenAI** to automate document matching, summarization, and personalized recommendations
- Designed evaluation metrics (embedding similarity, keyword overlap) to assess recommendation quality and improve alignment between resumes and job requirements
- Extended the system to recommend relevant learning resources (e.g., Coursera courses) to help users address identified skill gaps

Facial Expression Recognition | CNN, NumPy 2024

- Co-implemented CNN from scratch using NumPy to understand convolutional architectures
- Achieved 33.6% test accuracy on <5% of RAF-DB - analyzed performance gap relative to full-dataset baselines (85%) and attributed limitations to data scarcity and model capacity constraints

PUBLICATIONS

Y. Zhao, **R. Ravi**, et al. "PATE: Property, Amenities, Traffic & Emotions Coming Together for Real Estate Price Prediction" *2022 IEEE 9th Intl. Conf. on Data Science and Advanced Analytics (DSAA)*

Y. Zhao, S. Shi, **R. Ravi**, et al. "H4M: Heterogeneous, Multi-source, Multi-modal, Multi-view, Multi-distributional Dataset for Socioeconomic Analytics in Beijing" *2022 IEEE 9th Intl. Conf. on Data Science and Advanced Analytics (DSAA)*

PERSONAL

CFA Level 1, PADI Advanced Open Water Diver, Drexel Finance and Investment Group, HKUST International Students Soccer Team